

Om Kolhe

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Skills: IC Design, Nanofabrication, in-vivo Electrophysiology & Ca²⁺ Imaging, Computational Neuroscience.

Summary

- Researcher with 5+ years of academic experience in neurotechnology, neuroscience and ASIC design.

Publications ([Google Scholar](#))

- Neural Delay Lines Synchronize Inter-Areal Motor Sequences. (*in prep*) 2025
- NanoNeedleNet: A multi-modal platform for 3D electrophysiology and 2P imaging. (*in prep*) 2025
- Blood-Catalyzed n-Doped Conducting Polymer for Modulation of Neuronal Membranes. (*in revision*) 2025
- A chronic multimodal platform for simultaneous EPhys and Ca²⁺ imaging. Transducers (*talk*) 2025

Conference Presentations

- **Posters** – COSYNE 2025, SfN 2024, Barrels 2024, Big10 Neuroscience 2024, Barrels 2023
- **Oral Presentations** – Transducers 2025

Patents

- 2D and 3D Neural Electrodes and Methods Thereof. US Patent #63/542,491 2024
- Integrated chip-scale electrophysiology and electrochemistry detection and amplifying apparatus, system, and method for use thereof. (provisional PURD-152/00 US 28593/710) 2024

Internship Experience

- **Summer Intern.** SONY, Atsugi Technology Center, Japan. 05/18-08/18
 - Characterized and benchmarked SONY's Tuner IC with competitor's ICs to identify areas for improvement.
 - Developed a Software Defined Radio (SDR) technology feature for the Tuner IC.

Work Experience

- **Neuroengineering and Neuroscience.** *Purdue University*, PhD Thesis. 01/21-Current
 - Developed and optimized scalable thin-film 3D electrodes to combine two-photon microscopy and volumetric electrophysiology recordings.
 - Investigated skilled volitional movement in motor and pre-motor areas through experimental and computational models. ([Model](#))
 - Designed a custom IC for simultaneous electrical and electrochemical recording to study the role of dopamine in attention and decision making.
 - Developed a chronic platform for chronic multimodal ECoG and two-photon imaging, enabling longitudinal behavior studies.
- **Multiphase Oscillator for 5G Receivers.** *IIT Bombay*, Master's Thesis. 01/19-07/20
 - Designed a low-power, low-phase-noise multiphase oscillator for N-Path filters, achieving an 8-phase clock by synchronizing an injection-locked ring VCO with an on-chip LC oscillator.
 - The designed circuit taped-out in UMC 65nm Low Leakage (UMC65-LL) technology node
- **RF Receiver for IRNSS.** *IIT Bombay*, Undergraduate Research. 05/17-05/19
 - Developed a functional receiver for the Indian Regional Navigation Satellite System (IRNSS/NavIC) by designing and testing an S-band antenna and RF front-end circuitry for signal processing.

Education

- **PhD, Biomedical Engineering.** *Purdue University, Indiana.* Advisor: *Prof. Krishna Jayant* 2021-present
- **B.Tech + M.Tech in Microelectronics.** *IIT Bombay, India.* Advisor: *Prof. Maryam Baghini* 2015-2020

Skills

- **Software:** Python, Cadence Virtuoso, MATLAB, Verilog, VHDL, C++, Embedded C, KLayout, Suite2p, DeepLabCut, HFSS, ADS, Altium, EAGLE.
- **Fabrication:** Polymer based nanofabrication, Nanoscale 3D printing, Device characterization, mixed-signal PCB design.
- **Experimental:** Animal neurosurgery, ECoG and silicon probe recording, 2P calcium imaging.